

Partial Result: Complemented ℓ_p Subspaces Give Singular Extensions

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Source Problem

The source paper is Spiros A. Argyros, Antonis Manoussakis, and Pavlos Motakis, *The HI extension of the standard HI spaces*, arXiv:2407.20030 [1]. In its final discussion, the paper states W. B. Johnson's singular-extension problem. In the parsed source this is `data/parsed/arxiv_sources/2407.20030/source.tex`, lines 2026–2037. On PDF page 25, Definition 10.1 defines a singular extension, and Problem 3 asks whether every infinite-dimensional separable Banach space not isomorphic to c_0 admits one.

10. FINAL DISCUSSION

We discuss some open problems that are related to this article. We begin with the one that motivated this paper and remains open in its full generality.

Problem 1 (Pełczyński). *Does every HI space X embed into an HI space Y such that Y/X is infinite dimensional?*

Problem [1](#) was posed by Pełczyński at a conference where the first author was present, however we were unable to find it in the literature. Yet, he posed a strongly related one in [\[17\]](#) Problem 3.3, page 348].

Problem 2 (Pełczyński). *Does every separable HI space X embed into an HI space Y with a Schauder basis?*

It is remarkable that in the same paper [\[17\]](#) Pełczyński posed the problem of the existence of a $\mathcal{L}_\infty$ -space that is HI, a fact that the first author was unaware of. This is why this paper was not cited in [\[2\]](#), where the scalar-plus-compact problem was solved using such a space. Let us point out that the methods developed in the current paper show promise in constructing HI extensions of $\mathcal{L}_\infty$ -HI spaces, such as the Argyros-Haydon space.

After sharing a preliminary version of this article to W. B. Johnson, he pointed out to us a more general form of Pełczyński's HI extension Problem [1](#) based on the following definition (see also [\[11\]](#) page 4701] for a formulation in the setting of twisted sums).

Definition 10.1. *Let X be a Banach space Y be a superspace of X . We call Y a singular extension of X if Y/X is infinite dimensional and the quotient map $Q : Y \rightarrow Y/X$ is strictly singular.*

Problem 3 (W. B. Johnson). *Does every infinite dimensional separable Banach space X that is not isomorphic to c_0 admit a singular extension Y ?*

For example, as proved by Kalton and Peck, the space $X = \ell_p$, $1 < p < \infty$, admits a singular extension, namely the Banach space Z_p (see [\[16\]](#) Theorem 6.4, page 27]). It is not hard to see that if X is HI and it admits a singular extension Y then Y must be HI as well. Therefore, a positive answer to Problem [3](#) would imply a positive answer of Problem [1](#) at least in the separable setting (whether every separable HI space admits a singular extension was also asked by Castillo, Ferenczi, and Gonzalez in [\[11\]](#) page 4701]).

In [\[11\]](#) it is proved that the uniformly convex HI space $\mathcal{F}$ constructed by V.

Definitions

Let X be a Banach space and let Y be a superspace of X . Following the source paper, Y is a *singular extension* of X if Y/X is infinite dimensional and the quotient map

$$Q : Y \rightarrow Y/X$$

is strictly singular. Equivalently, Q is not an isomorphism on any infinite-dimensional subspace of Y .

A subspace E of a Banach space X is complemented if $X = E \oplus F$ for some closed subspace F , or equivalently if there is a bounded projection from X onto E .

Idea of the Proof

Singular extensions are stable under adding an external complemented summand. If E already has a singular extension $0 \rightarrow E \rightarrow Z \rightarrow V \rightarrow 0$, then $X = E \oplus F$ is extended by $Y = Z \oplus F$. The quotient still only sees the Z -coordinate. Thus, any infinite-dimensional subspace on which the new quotient map is bounded below would project injectively to an infinite-dimensional subspace of Z , where the old quotient map is also bounded below. That contradicts the singularity of the original extension.

Complemented-Summand Permanence

Partial Theorem. *Let X be a Banach space containing a complemented subspace E . If E admits a singular extension, then X admits a singular extension.*

Proof. Choose a closed complement F so that $X = E \oplus F$. Let

$$0 \longrightarrow E \xrightarrow{\iota} Z \xrightarrow{q} V \longrightarrow 0$$

be a singular extension of E . Thus V is infinite dimensional, $\ker q = \iota(E)$, and q is strictly singular.

Let $Y = Z \oplus_1 F$. Embed $X = E \oplus F$ into Y by

$$J(e + f) = (\iota e, f).$$

Define

$$\tilde{q} : Y \rightarrow V, \quad \tilde{q}(z, f) = qz.$$

Then $\tilde{q}$ is a bounded surjection and $\ker \tilde{q} = J(X)$. Hence $Y/J(X)$ is isomorphic to V , so the quotient is infinite dimensional.

It remains to show that $\tilde{q}$ is strictly singular. Suppose not. Then there is an infinite-dimensional subspace $M \subset Y$ and a constant $c > 0$ such that

$$\|\tilde{q}y\| \geq c\|y\| \quad (y \in M).$$

Let $P_Z : Y \rightarrow Z$ be the coordinate projection. If $y \in M$ and $P_Z y = 0$, then $\tilde{q}y = 0$, hence $y = 0$. Thus $P_Z|_M$ is injective, and $P_Z M$ is infinite dimensional. For $z = P_Z y \in P_Z M$,

$$\|qz\| = \|\tilde{q}y\| \geq c\|y\| \geq c\|P_Z y\| = c\|z\|,$$

where the inequality $\|y\| \geq \|P_Z y\|$ uses the ℓ_1 -sum norm. Therefore q is bounded below on the infinite-dimensional subspace $P_Z M$, contradicting that q is strictly singular. Hence $\tilde{q}$ is strictly singular, and Y is a singular extension of X . $\square$

Corollary. *Let X be a separable Banach space. If X contains a complemented copy of ℓ_p for some $1 < p < \infty$, then X admits a separable singular extension.*

Proof. Kalton and Peck proved that ℓ_p , $1 < p < \infty$, admits a singular extension Z_p [2, Theorem 6.4]; the source paper recalls this example immediately after Johnson's Problem 3. Transport this extension across the isomorphism between the complemented subspace of X and ℓ_p , and apply the theorem. Separability is preserved because X , the complement, and the Kalton–Peck extension Z_p are separable. $\square$

Scope and Limitations

This gives a positive answer to Johnson’s Problem 3 for the concrete class of separable Banach spaces containing a complemented copy of ℓ_p , $1 < p < \infty$. It is not a solution of Johnson’s full problem, since it does not cover arbitrary separable spaces not isomorphic to c_0 . It is also not a solution of Pelczynski’s HI-extension problem, because HI spaces do not decompose into nontrivial complemented infinite-dimensional summands.

Novelty and Search Bounds

The run indexes were searched for arXiv:2407.20030, HI extension, singular extension, Kalton–Peck, Z_p , pushout, pullback, twisted sum, and strictly singular quotient-map terminology. No prior packet for this source paper or for this complemented-summand permanence lemma was found. The result is a small but useful permanence observation, not a claimed new construction of singular extensions beyond the complemented-subspace mechanism.

Human Review Recommendation

Check the permanence lemma against the exact convention for singular twisted sums. The proof uses the quotient-map definition from the source paper and the ℓ_1 -direct sum only to make the projection estimate immediate.

References

- [1] S. A. Argyros, A. Manoussakis, and P. Motakis, *The HI extension of the standard HI spaces*, arXiv:2407.20030, 2024.
- [2] N. J. Kalton and N. T. Peck, *Twisted sums of sequence spaces and the three space problem*, Trans. Amer. Math. Soc. 255 (1979), 1–30.